

Meheru Zannat

AI/ML Engineer

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Computer Science graduate (KUET) and published ML researcher who designs and extends **deep-learning architectures** in **Python / PyTorch** — **CNNs**, **Transformers** with cross-attention, and **latent diffusion** models — and ships them as production **LLM systems**: **RAG** pipelines over **vector databases** and **LangChain / LangGraph** tool-calling agents, served through **FastAPI**, **Docker** and **CI/CD** on cloud platforms.

EDUCATION

Khulna University of Engineering & Technology (KUET)

B.Sc. in Computer Science & Engineering

Khulna, Bangladesh

Mar 2022 – Jun 2026

RESEARCH & PUBLICATIONS

Zannat, M. “[Parkinson’s Disease Detection via Self-Supervised Dual-Channel Cross-Attention on Bilateral Wrist-Worn IMU Signals.](#)” arXiv:2604.18372, 2026.

- Designed a **self-supervised dual-channel cross-attention** encoder over bilateral wrist-IMU signals (PADS, 469 subjects): **93.12% accuracy**, rising to **93.56% with only 20% of labels** after **contrastive (InfoNCE) pretraining** and fine-tuning — then **optimised for edge inference** at **48 ms/window** on a Raspberry Pi.

Undergraduate Thesis — Tropical Cyclone Forecasting via Latent Rectified Flow.

KUET CSE

- Built a two-stage **latent rectified-flow** model (multi-channel VAE + conditional UNet with **factorized temporal attention**) that **jointly forecasts** satellite infrared imagery and four ERA5 atmospheric fields in a **single pass** — **16.35 dB PSNR / 0.759 SSIM**, beating a cascaded-diffusion baseline at every lead time while sampling **~30× faster** (56 ms vs 1673 ms).
- Read the storm track physically out of the generated winds via a **steering-flow** calculation (no track head): **62.4 km** error at +9 h, 15% below baseline; **reward fine-tuning (DRAFT)** through the sampler cut it a further **8–11%**.

PROJECTS

AI Chatbot — Retrieval-Augmented Generation & Agentic LLM System

[Live](#) · [Code](#)

- Built a production **RAG** assistant — OpenAI embeddings indexed in a **FAISS** vector store — on a **FastAPI** backend with SSE token streaming, plus a **LangGraph** tool-calling agent (Wikipedia, arXiv) that plans and executes multi-step responses.
- Shipped the full stack (React client + Python inference service) to cloud hosting, with integration tests around the streaming API.

Braillelife — Multimodal Vision + Language Recognition

[Demo](#) · [Code](#)

- Built an end-to-end **multimodal** pipeline: an **RF-DETR** object detector (self-annotated dataset, trained and served via Roboflow) feeding a **LangChain + OpenAI** agent that corrects detections, scores confidence and answers follow-up questions through tool lookup.
- Deployed the system as **3 independently deployable inference microservices** on serverless cloud hosting, with a companion Flutter Android client.

Banchan — Event-Driven Microservices Platform

[Live](#) · [Code](#)

- Architected **6 Node.js/Express microservices** behind an API gateway (JWT auth, Redis-backed rate limiting, database-per-service on MongoDB with cache-aside) and an **event-driven** BullMQ/Redis pipeline with retries and exponential backoff; covered by **Jest/Supertest** suites on a **GitHub Actions** CI matrix and **Docker**-based cloud deployment.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript / TypeScript, Java, Bash

ML Frameworks & Libraries: PyTorch, TensorFlow / Keras, Hugging Face, scikit-learn, OpenCV, NumPy, Pandas

Architectures & Methods: CNNs, Transformers & attention, encoder-decoder, **diffusion models**, self-supervised & contrastive learning, pretraining & **fine-tuning**, RF-DETR, YOLO, model optimisation for edge inference

LLM & Agentic AI: LangChain, LangGraph, **RAG**, FAISS / vector databases, OpenAI & Groq APIs, function & tool calling, prompt engineering

MLOps & Cloud: Docker, CI/CD (GitHub Actions), Git, Linux / Bash, Amazon Web Services (AWS), model serving (FastAPI, Roboflow), Hugging Face Hub, Vercel, Render

Backend & Data: FastAPI, Node.js / Express, REST API design, microservices; PostgreSQL, MySQL, MongoDB, Redis

ACHIEVEMENTS

Competitive Programming: [Codeforces](#) (120+ solved) · [LeetCode](#) (60+ solved) — data structures & algorithms.